

# **Goal: Create YAL.ar's first real-world fraud dataset pipeline → foundation for AI fraud detection at scale.**

## **Introduction / Context :**

Fraud is one of the biggest risks for AI-powered systems. Toy datasets exist, but they fail to capture the messy, adversarial, and multilingual nature of real-world fraud. This project aims to bridge that gap by building a scalable pipeline to collect, clean, and structure fraud conversations from open platforms such as Twitter/X and Reddit.

The outcome will be a dataset that supports YAL.ai in training robust fraud detection models.

Why this is important:

- Fraud is a big risk for any AI system.
- To stop fraud, we need real-world data that shows how fraud happens.
- With this project, we will create a pipeline to collect, clean, and organize this data so it can be used by AI models in the future.

## **Objectives :**

The main objectives of this project are:

1. To design a pipeline that collects and organizes fraud-related data.
2. To make sure the data is clean, correct, and useful for AI models.
3. To create a system that can handle large amounts of data in the future.
4. To support YAL.ai in building a fraud detection system that is reliable and scalable.

## **Target Fraud Categories :**

We will focus on multiple fraud types to ensure diversity:

- Lottery & Prize scams (e.g., “You won 1 lakh, send details”)
- KYC / Banking scams (fake verification links)
- Crypto scams (investment & pump-and-dump groups)
- Job scams (fake recruiter messages)
- Loan/Financial scams (instant loan, hidden charges)
- Phishing attempts (urgent password reset, OTP sharing)

## **Platforms Selected & Justification :**

- Twitter/X – Rich in trending scam discussions, scammer outreach attempts, hashtags (#KYC, #crypto, #jobscam). Public, multilingual content.
- Reddit – Structured threads around fraud (“r/scams”, “r/india”), detailed discussions and victim conversations. Easy to scrape using APIs.

## **Approach / Phases :**

### **Phase 1: Research**

- Identify fraud categories & keywords.
- Collect sample fraud posts manually to guide scraping logic.

### **Phase 2: Pipeline Design**

- Tools: Python 3.10+, **snsrape** for Twitter, **praw** for Reddit.
- Schema: **{raw\_text, platform, fraud\_type, language, label}**.
- Store in CSV/JSON.

### Phase 3: Data Collection

- Scrape 200+ fraud samples (minimum 100 required).
- Ensure full conversations (thread replies, Reddit comment chains).

### Phase 4: Cleaning & Annotation

- Remove duplicates, irrelevant posts.
- Manually annotate 20–30 samples for validation.
- Check multilingual coverage.

### Phase 5: Scaling Strategy

- Parallelize scraping scripts for higher volume.
- Store in databases (MongoDB/Postgres) instead of flat files.
- Deploy on cloud (AWS/GCP) for continuous pipeline runs.

## Approach / Phases :

### Ethics:

- We will only use public data from platforms.
- No private or personal information will be collected.

- **Data will be anonymized and used responsibly.**

**Multilingual Consideration:**

- **Data will include posts in English and Hindi.**
- **Keywords like “urgent”, “KYC”, “lottery” will be considered in both languages.**